Kushal Mundra

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Summary

MSc Computer Science graduate (Distinction, University of Warwick) who builds Generative AI and Agentic AI applications in Python: a deployed Claude-powered conversational AI assistant with tool calling and API integrations, and a RAG pipeline that grounds LLMs in a knowledge graph, with outputs tested and validated against defined metrics. Three peer-reviewed publications.

Education

University of Warwick Sept. 2025 – Sept. 2026
MSc Computer Science (Distinction) *Coventry, UK*
Modules: Machine Learning & AI, NLP, Data Analytics, High Performance Computing, Image & Video Analysis

FLAME University Aug. 2021 – May 2024
BSc (Hons.) Computer Science *Pune, India*

Experience

AI Center, FLAME University April 2025 – July 2025
Software Engineering Intern *Pune, India*

- Built and deployed Kira, a Claude-powered conversational AI assistant for live meetings (Anthropic Claude 3.5 Sonnet), running a real-time, event-driven Python pipeline over WebRTC (LiveKit) with Deepgram speech-to-text and Resemble text-to-speech.
- Implemented agentic tool calling for web search and calendar management through REST APIs, and gave the assistant meeting context by logging the transcript and injecting it into the prompt at query time.
- Tested the assistant in live sessions and fixed a turn-handling issue by moving wake-word gating ahead of the LLM call, which stopped replies to every utterance and cut unnecessary API calls and cost.

Indian Institute of Technology Bombay Sept. 2024 – Feb. 2025
Project Technical Assistant *Mumbai, India*

- Evaluated 6 AI coding assistants and benchmarked the top 2 (Amazon Q, Tabnine) across 9 contests on LeetCode, CodeChef and Codeforces, analysing correctness and code quality.
- Designed a rubric-based evaluation framework for AI-generated output, led a user study with 18 participants and documented findings; co-authored paper at AIED 2025 (Springer).

Projects

PersonaGraph: RAG for Grounded LLM Generation | *Python, LLMs, RAG, Knowledge Graph, Ollama* 2026

- MSc dissertation: prepared and validated a knowledge source from 4 government PDF reports, generating 500 synthetic records and a knowledge graph (950 nodes, 6 entity types) for retrieval.
- Built a RAG pipeline that retrieves each record's graph context into the prompt for open LLMs (Qwen 3.5, Gemma 4, Mistral); consistency rose from 9.5% to 71.0% (Wilcoxon $p < 0.001$), with a plain-text baseline as an ablation.
- Tested 5 models x 100 cases on a GPU cluster, compared zero-shot, few-shot and chain-of-thought prompting, and validated outputs with BERTScore and NLI, finding the first metric overstated quality by about 20 pp.

Tweet Sentiment Classification (SemEval 2017) | *Python, PyTorch, Hugging Face, Scikit-learn* 2025

- Fine-tuned a RoBERTa Transformer and compared it with 6 other classifiers; the best model achieved a macro F1 of 0.8552 across three test sets.

Publications

- O. Joshi, **K. Mundra** et al., "AI Coding Assistants in Competitive Programming: Empirical Studies on Human-AI Interactions," *AIED 2025*, Springer CCIS vol. 2590.
- A. Varsha, **K. Mundra** et al., "From Pixels to Insight: Enhancing Metallic Component Defect Detection with GLCM Features and AI Explainability," *ICDMAI 2024*, Springer LNNS vol. 997.
- **K. Mundra** et al., "Fostering Ethics in AI: Perceptions from the Indian AI Curriculum," *ICCE 2023*, APSCE.

Technical Skills

GenAI & Agents: Generative AI, LLM Applications, Prompt Engineering, RAG, Agentic AI, Tool Calling, Workflow Orchestration, LangChain, LangGraph, Conversational AI

ML & Evaluation: Machine Learning, Scikit-learn, PyTorch, TensorFlow, Hugging Face, Embeddings, Vector Databases, Functional Testing & Validation

Programming & APIs: Python (Advanced), SQL, JSON, REST APIs, Git, Docker, CI/CD, Linux, Technical Documentation, SDLC

Cloud & Platforms: Microsoft Azure (Azure OpenAI), AWS (Bedrock, SageMaker), Anthropic Claude, Ollama, Enterprise Knowledge Retrieval